Lab Session 04

Explore time parameter and classify discrete time system

Manipulation of the time parameter

In this Lab we will look at three signal operations affecting the time parameter of the signal; time shifting, time scaling and time reversal. These operations are very common components to real-world systems and, as such, should be understood thoroughly when learning about signals and systems.

Some common operations on signals affect the time parameter of the signal. One of these is time shifting in which a quantity is added to the time parameter in order to advance or delay the signal. Another is the time scaling in which the time parameter is multiplied by a quantity in order to dilate or compress the signal in time. In the event that the quantity involved in the latter operation is negative, time reversal occurs.

Time shifting:

Time shifting is, as the name suggests, the shifting of a signal in time. This is done by adding or subtracting a quantity of the shift to the time variable in the function. Subtracting a fixed positive quantity from the time variable will shift the signal to the right (delay) by the subtracted quantity, while adding a fixed positive amount to the time variable will shift the signal to the left (advance) by the added quantity. As shown in figure 4.1 , f (t−T) moves f (t) to the right by T which is an example of delaying a signal.

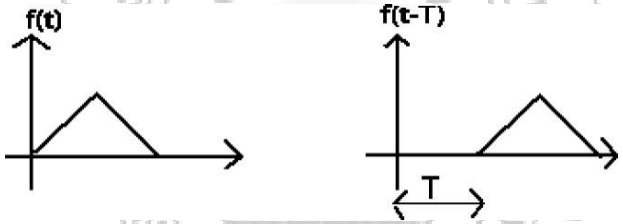


Figure. 4.1: *Time shifting of a signal*

If a signal can be expressed in the form as shown below in equation 4.1, we say that x(t) is a *time-shifted* version of s(t). An example of time shifting is also shown in figure 4.2 for graphical illustration.

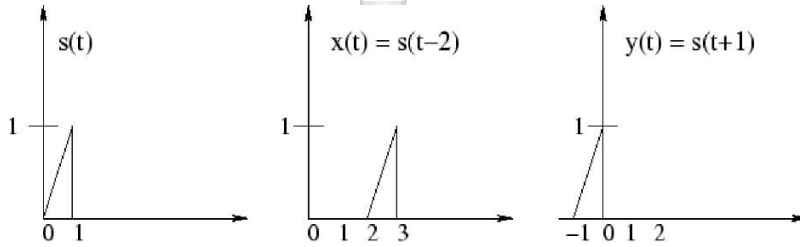


Figure. 4.2: *Time-shifting of a signal **s**(t)*

Consider a simple function as shown in equations 4.1 below, when this function is shifted by $t_1=2$ seconds it gives the equation 4.2 which is simply s(t) with its origin shifted to the right, or **delayed**, by 2 seconds.

$$\mathbf{s(t) = t} \qquad \mathbf{0 \le t \le 1} \qquad \mathbf{(4.1)}$$
$$\mathbf{x(t) = s(t-2) = t - 2} \qquad \mathbf{0 \le t-2 \le 2}$$

$$= t - 2 \qquad 2 \le t \le 3 \qquad (4.2)$$

Similarly, shifting the function s(t) as shown in equation, by $t_1 = -1$ seconds yields equation 3 which is simply s (t) with its origin shifted to the left, or **advance in time**, by 1 second.

$$y(t) = s(t+1) = t + 1 \qquad 0 \le t + 1 \le 1$$
$$= t + 1 \qquad -1 \le t \le 0 \qquad (4.3)$$

A positive phase indicates a shift to the left whereas a negative phase indicates a shift to the right.

Time scaling:

Time scaling compresses or dilates a signal by multiplying the time variable by some quantity. If that quantity is greater than one, the signal becomes narrower and the operation is called compression, while if the quantity is less than one, the signal becomes wider and is called dilation.

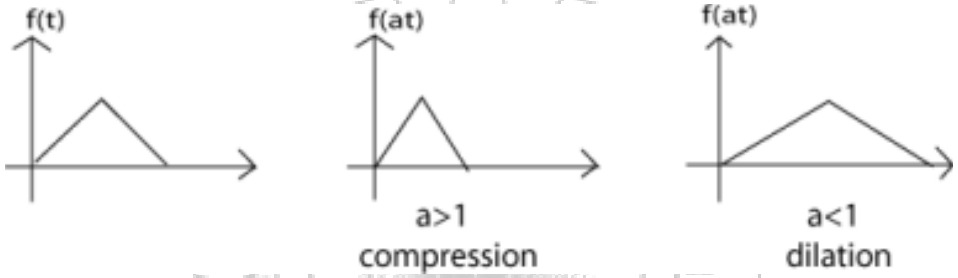


Figure 4.3: *Time scaling of a signal 'f(t)'*

Time reversal:

A natural question to consider when learning about time scaling is: What happens when the time variable is multiplied by a negative number? The answer to this is time reversal. This operation is the reversal of the time axis, or flipping the signal over the y-axis.

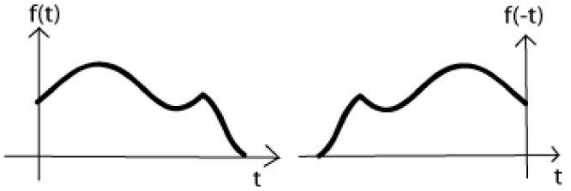


Figure 4.4: *Reversal of the time axis*

Classification of discrete time systems

Discrete systems & its properties

A discrete-time system takes a sequence x(n) called excitation and transforms it into another sequence y(n) (called response). A system which possesses both Linearity and Time invariance properties is known as LTI system.

1- linearity

The two conditions of linearity are additive rule and homogeneity rule shown in eq. 4.4 and eq. 4.5 respectively. These two criteria can be combined into one equation known as the '*superposition principle*' is shown in equation 4.6 and depicted in Figure 4.5.

$$\mathbf{y[n] = Sys\ \{\ x_1\ [n] + x_2\ [n]\ \} = Sys\ \{\ x_1\ [n]\ \} + Sys\ \{\ x_2\ [n]\ \}} \quad \mathbf{(4.4)}$$

$$\mathbf{y[n] = Sys\ \{\ \alpha\ x_1\ [n]\ \} = \alpha\ Sys\ \{\ x_1\ [n]\ \}} \quad \mathbf{(4.5)}$$

$$\mathbf{y[n] = Sys\ \{\ \alpha_1\ x_1\ [n] + \alpha_2\ x_2\ [n]\ \} = \alpha_1\ Sys\ \{\ x_1\ [n]\ \} + \alpha_2\ Sys\ \{\ x_2\ [n]\ \}} \quad \mathbf{(4.6)}$$

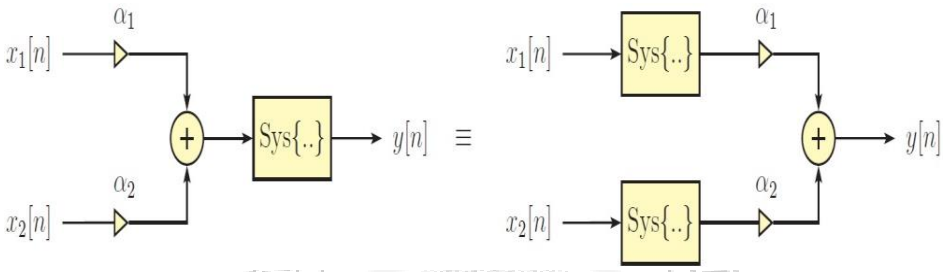


Figure 4.5: *Diagrammatic representation of Linearity property*

2- time invariance

A system is called time invariant if its input-output characteristics do not change with time and a system whose impulse response h(n) doesn't change with respect to time. The Condition for time invariance is shown in eq 4.7. For the system to be considered time-invariant, the only effect of time-shifting the input signal should be to cause an equal amount of time shift in the output signal shown in Figure 4.6.

$$\mathbf{Sys\ \{\ _1\ [n]\}\ = y[n] \qquad implies\ that \qquad Sys\ \ x_1\ [n - k] = y[n - k]} \quad \mathbf{(4.7)}$$

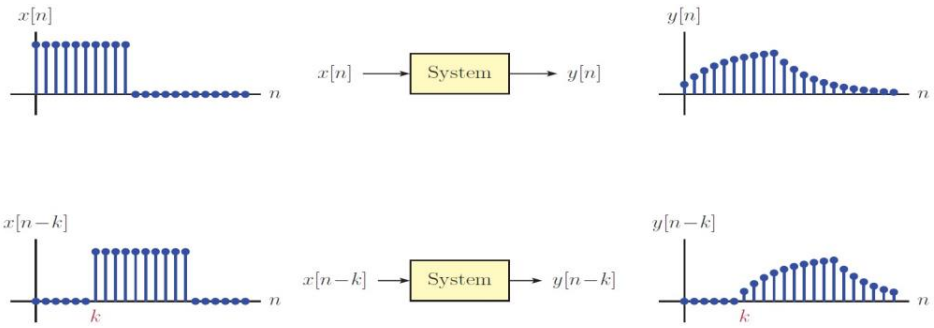


Figure 4.6: *Diagrammatic representation of Time Invariance property*

EXERCISES:

Task # 1: Write a script that delays the unit impulse by 5 samples.

Task # 2: Generate a script that uses user input to shift unit impulse right or left. Use MATLAB input command to take the number of samples to shift the signal and use switch option or If - else statement to decide whether to shift the signal to the right (delay) or to the left (advance).

Task # 3: Repeat task # 2 with unit step response.

Task # 4: Create a function that flip the sequence first and then advance the sequence. The function will receive two arguments as defined below, first is input signal which needs to be flipped and then advances, and second is the number of samples by which the signal should be advanced.

[x]= *flipadv* (u,s)

Task # 5: Generate a function that advances the sequence first and then flip the sequence. The function will receive two arguments as defined below, first is input signal which needs to be moved and then flipped, and second is the number of samples by which the signal should be advanced.

[x]= *advflip* (u,s)

Task # 6: Create a Triangular wave with unit amplitude. Use subplot command to show the effect of compression and dilation.

Task # 7: Generate a Square wave with amplitude from -5 to 5. Plot the signal which shows the original signal along with its flipped signal.

Task # 8: Prove the Linearity property by using the following parameters

a) x1 = Unit Step (Step time = 1 & Final value = 1)
b) x2 = Unit Step (Step time = 5 & Final value = 3)
c) $\alpha 1 = 1$
d) $\alpha 2 = 2$
e) Sys(H) = x (n – 3)

Task # 9: Verify the Linearity property by using the following parameters

a) x1 = Unit Step (Step time = 1 & Final value = 1)
b) x2 = Unit Step (Step time = 5 & Final value = 3)
c) $\alpha 1 = 7$
d) $\alpha 2 = 19$
e) Sys(H) = 5 * X (n – 7)

Task # 10: Check whether the system and whose parameters are as follows, is Linear or not?

a) x1 = Unit Step (Step time = 1 & Final value = 1)
b) x2 = Unit Step (Step time = 5 & Final value = 3)
c) $\alpha 1 = 7$
d) $\alpha 2 = 19$
e) Sys(H) = $X^2(n-1)$

Task # 11: Check the property of Time Invariance on systems of task 1 to task 4. Also comment whether the system is LTI or not?

Task # 12: By using hand calculations, prove that the results generated by SIMULINK are correct. Check whether the systems of task 1 to task 4 are LTI or not?

Task # 13: Prove Linearity property by giving random signal and system as an input.

Task # 14: Design Time Invariance System by giving random signal and system.

NED UNIVERSITY OF ENGINEERING & TECHNOLOGY
KARACHI